

Solution to Ramanujan Challenge Problem 2.3: The Sum $\pi + e$ as an Apéry Limit

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Abstract

We prove that the order-four recurrence of Problem 2.3 converges to $\pi + e$. The mechanism is that the challenge operator is a *tensor product*: order $4 = 2 \times 2$. It annihilates every product $X_{n+2}Y_{n+3}$ of a solution X of Lambert's continuant recurrence (whose ratio produces $\pi/4$) with a solution Y of the derangement recurrence (whose ratio produces e). The key observation is that $m!$ satisfies the *same* recurrence as the derangement numbers D_m ; this is why π and e appear together, and additively. The challenge's own p_n, q_n are then identified with explicit tensor solutions, and the ratio splits *exactly*, with no asymptotic analysis:

$$\frac{p_n}{q_n} = 4 \frac{B_{n+2}}{A_{n+2}} + \frac{(n+3)!}{D_{n+3}}.$$

Every step is proved here and machine-checked in Lean 4. In particular, the Lambert limit $B_m/A_m \rightarrow \pi/4$ is derived from a positive moment integral and a geometric remainder bound; it is not cited as an external input.

1 The problem

For $n \geq 1$ the challenge poses the recurrence

$$\begin{aligned} 0 = & (-n^3 + 2n^2 + 7n + 3) u_n \\ & + (n+2)(2n^4 + n^3 - 26n^2 - 48n - 19) u_{n-1} \\ & + (n+2)(n^6 + 9n^5 + 8n^4 - 87n^3 - 249n^2 - 234n - 68) u_{n-2} \\ & + (n+1)^2(n+2)(2n^5 + 3n^4 - 13n^3 - 21n^2 + 4) u_{n-3} \\ & - n^3(n+1)^2(n+2)(n^3 + n^2 - 8n - 11) u_{n-4}, \end{aligned} \tag{1}$$

and asks for $\lim_{n \rightarrow \infty} p_n/q_n = \pi + e$, where p, q are the solutions with

$$(p_{-3}, p_{-2}, p_{-1}, p_0) = (1, 1, 20, 296), \quad (q_{-3}, q_{-2}, q_{-1}, q_0) = (1, 0, 4, 48). \tag{2}$$

Write $c_0(n), \dots, c_4(n)$ for the five coefficient polynomials in (1), in the order displayed, so that (1) reads $\sum_{j=0}^4 c_j(n) u_{n-j} = 0$.

2 Two order-two systems

Definition 2.1. The Lambert system $\mathcal{L}$ is the recurrence

$$X_m = (2m+1)X_{m-1} + m^2 X_{m-2} \quad (m \geq 1), \quad (3)$$

and the derangement system $\mathcal{D}$ is

$$Y_m = (m-1)(Y_{m-1} + Y_{m-2}) \quad (m \geq 2). \quad (4)$$

Distinguished solutions: A_m, B_m of $\mathcal{L}$ with $A_{-1} = A_0 = 1$ and $B_{-1} = 0, B_0 = 1$; and D_m (the derangement numbers) of $\mathcal{D}$ with $D_0 = 1, D_1 = 0$.

Thus $A_1 = 4, A_2 = 24, A_3 = 204; B_1 = 3, B_2 = 19, B_3 = 160$; and $D_2 = 1, D_3 = 2, D_4 = 9$.

Lemma 2.2 (The factorial is a second solution of $\mathcal{D}$). $F_m := m!$ satisfies (4).

Proof. $(m-1)((m-1)! + (m-2)!) = (m-2)!(m-1)((m-1)+1) = (m-2)!(m-1)m = m!$. $\square$

Lemma 2.2 is the structural reason the constant e occurs in this problem, and the reason it occurs *additively* alongside π : the $\mathcal{D}$ -system carries both D_m and $m!$, and their ratio is the classical approximation to e .

Lemma 2.3 (Casorati determinants). For all $m \geq 0$,

$$A_m B_{m-1} - A_{m-1} B_m = (-1)^{m+1} (m!)^2, \quad D_m F_{m-1} - D_{m-1} F_m = (-1)^m (m-1)! \quad (m \geq 1).$$

In particular $\{A, B\}$ is a basis of the solution space of $\mathcal{L}$ and $\{D, F\}$ is a basis of the solution space of $\mathcal{D}$.

Proof. Write $W_m = A_m B_{m-1} - A_{m-1} B_m$. Substituting (3) for A_m and B_m , the $(2m+1)$ -terms cancel and $W_m = -m^2 W_{m-1}$; with $W_0 = A_0 B_{-1} - A_{-1} B_0 = -1$ this telescopes to $W_m = (-1)^{m+1} (m!)^2$. The second identity is the same computation: substituting (4) gives $D_m F_{m-1} - D_{m-1} F_m = -(m-1)(D_{m-1} F_{m-2} - D_{m-2} F_{m-1})$, and the initial value at $m = 1$ is $D_1 F_0 - D_0 F_1 = -1$. $\square$

3 The tensor theorem

Theorem 3.1 (Tensor annihilation). Let X be any solution of $\mathcal{L}$ and Y any solution of $\mathcal{D}$. Then

$$u_n := X_{n+2} Y_{n+3}$$

satisfies the challenge recurrence (1) for every $n \geq 1$.

Proof. The assertion is an identity in the four free initial values X_{-1}, X_0, Y_0, Y_1 , so it suffices to prove a polynomial identity. Put

$$\mathbf{a}(m) = \begin{pmatrix} 2m+1 & m^2 \\ 1 & 0 \end{pmatrix}, \quad \mathbf{d}(m) = \begin{pmatrix} m-1 & m-1 \\ 1 & 0 \end{pmatrix},$$

so that $(X_m, X_{m-1})^\top = \mathbf{a}(m)(X_{m-1}, X_{m-2})^\top$ and $(Y_m, Y_{m-1})^\top = \mathbf{d}(m)(Y_{m-1}, Y_{m-2})^\top$. Then the vector

$$v_n := (X_{n+2}, X_{n+1})^\top \otimes (Y_{n+3}, Y_{n+2})^\top \in \mathbb{Q}^4$$

obeys $v_n = T(n) v_{n-1}$ with $T(n) = \mathbf{a}(n+2) \otimes \mathbf{d}(n+3)$, and $u_n = e_1^\top v_n$ where e_1 is the first standard covector. Hence

$$\sum_{j=0}^4 c_j(n) u_{n-j} = \left(\sum_{j=0}^4 c_j(n) e_1^\top T(n) T(n-1) \cdots T(n-j+1) \right) v_{n-4},$$

the empty product being the identity. As the initial data vary, v_{n-4} ranges over a spanning set of $\mathbb{Q}^4$ (Lemma 2.3 supplies four independent products), so the recurrence holds for all solutions if and only if the covector in parentheses vanishes identically — that is, if and only if four polynomial identities in $\mathbb{Z}[n]$ hold.

Concretely, one eliminates X_{n+2}, X_{n+1}, X_n and $Y_{n+3}, Y_{n+2}, Y_{n+1}$ using (3) and (4) until $\sum_j c_j(n) u_{n-j}$ is a bilinear form in the four products

$$X_{n-2}Y_{n-1}, \quad X_{n-2}Y_n, \quad X_{n-1}Y_{n-1}, \quad X_{n-1}Y_n,$$

whose four coefficients are polynomials in n . All four are identically zero.

This is a finite computation in $\mathbb{Z}[n]$, and it has been carried out three independent ways: (i) in Lean 4, where after substitution it is discharged by the `ring` normaliser and is therefore kernel-checked (`RamanujanChallenge.P23.tensor_rec`); (ii) symbolically over $\mathbb{Q}[n]$ with four free initial values; and (iii) in exact integer arithmetic for the concrete sequences. See §6 and §7. $\square$

Corollary 3.2. *The solution space of (1) over $\mathbb{Q}$ is exactly four dimensional and is spanned by*

$$\{A_{n+2}D_{n+3}, A_{n+2}F_{n+3}, B_{n+2}D_{n+3}, B_{n+2}F_{n+3}\}.$$

Thus the challenge operator is the tensor product of $\mathcal{L}$ and $\mathcal{D}$, not merely annihilated by it.

Proof. By Lemma 2.3 the four listed sequences are linearly independent: their value matrix at two consecutive indices is the Kronecker product of two invertible 2×2 matrices. By Theorem 3.1 they solve (1). Since $c_0(n) \neq 0$ for $n \geq 1$ (Lemma 4.1 below), the solution space of (1) has dimension exactly four. $\square$

4 Identification of the challenge sequences

Lemma 4.1. $c_0(n) = -n^3 + 2n^2 + 7n + 3 \neq 0$ for every integer $n \geq 1$; consequently a solution of (1) is uniquely determined by $(u_{-3}, u_{-2}, u_{-1}, u_0)$.

Proof. For $n = 1, 2, 3, 4$ the values are 11, 17, 15, -1 . For $n \geq 5$, $n^3 - 2n^2 - 7n - 3 = n^2(n-2) - 7n - 3 \geq 3n^2 - 7n - 3 > 0$, so $c_0(n) < 0$. As $c_0(n) \neq 0$, equation (1) solves for u_n at each $n \geq 1$. $\square$

Proposition 4.2 (Closed forms). *The challenge sequences of (2) are*

$$q_n = A_{n+2} D_{n+3}, \quad p_n = 4 B_{n+2} D_{n+3} + A_{n+2} (n+3)!. \quad (5)$$

Proof. By Theorem 3.1 the sequence $A_{n+2}D_{n+3}$ solves (1); so do $B_{n+2}D_{n+3}$ and, by Lemma 2.2, $A_{n+2}(n+3)!$. Since (1) is linear, the right-hand sides of (5) are solutions. It remains to check four initial values of each, using $A_{-1} = A_0 = 1, A_1 = 4, A_2 = 24, B_{-1} = 0, B_0 = 1, B_1 = 3, B_2 = 19$ and $D_0 = 1, D_1 = 0, D_2 = 1, D_3 = 2$:

$$\begin{aligned} A_{-1}D_0 &= 1, & A_0D_1 &= 0, & A_1D_2 &= 4, & A_2D_3 &= 48; \\ 4B_{-1}D_0 + A_{-1}0! &= 1, & 4B_0D_1 + A_01! &= 1, & 4B_1D_2 + A_12! &= 20, & 4B_2D_3 + A_23! &= 296. \end{aligned}$$

These are exactly (2). By Lemma 4.1 the solution with prescribed initial data is unique, so (5) holds for all n . $\square$

Corollary 4.3 (Exact splitting). *For every $n \geq -1$,*

$$\frac{p_n}{q_n} = 4 \frac{B_{n+2}}{A_{n+2}} + \frac{(n+3)!}{D_{n+3}}. \quad (6)$$

Proof. Divide (5). The denominators are nonzero: $A_m > 0$ for all m by induction on (3) (both coefficients are positive for $m \geq 1$), and $D_m > 0$ for $m \geq 2$ likewise. $\square$

We stress that (6) is an identity, valid term by term; no asymptotic or Poincaré–Perron argument is used to obtain it. All that remains is to evaluate the two limits separately.

5 The two limits

Lemma 5.1 (The e half). $\lim_{m \rightarrow \infty} \frac{m!}{D_m} = e$.

Proof. Write $S_m := \sum_{k=0}^m (-1)^k / k!$. We claim $D_m = m! S_m$ for all $m \geq 0$. This holds at $m = 0$ ($1 = 1 \cdot 1$) and $m = 1$ ($0 = 1 \cdot (1 - 1)$). Assume it at $m-1$ and $m-2$. Using $(m-1)! = (m-1)(m-2)!$,

$$(m-1)(D_{m-1} + D_{m-2}) = (m-1)[(m-1)! S_{m-1} + (m-2)! S_{m-2}] = (m-1)! [(m-1)S_{m-1} + S_{m-2}].$$

Now $S_{m-2} = S_{m-1} - (-1)^{m-1}/(m-1)!$, so the bracket equals $m S_{m-1} - (-1)^{m-1}/(m-1)!$ and therefore

$$(m-1)(D_{m-1} + D_{m-2}) = m! S_{m-1} - (-1)^{m-1} = m! S_{m-1} + (-1)^m = m! \left(S_{m-1} + \frac{(-1)^m}{m!} \right) = m! S_m,$$

which is D_m . Hence $m!/D_m = S_m^{-1}$. The partial sums S_m converge to $e^{-1} \neq 0$, and $S_m > 0$ for $m \geq 2$ because an alternating series with strictly decreasing terms has all its partial sums from index 2 on trapped between $S_2 = 1/2$ and $S_3 = 1/3$. So the reciprocals converge to e . $\square$

Theorem 5.2 (The π half; Lambert). $\lim_{m \rightarrow \infty} \frac{B_m}{A_m} = \frac{\pi}{4}$.

Proof. Put

$$s = \sqrt{2}, \quad c = 2 - s, \quad P(x) = x(1 - x), \quad D(x) = 1 - cP(x),$$

and, for $r \geq 0$, define the positive moments

$$K_r := \int_0^1 \frac{P(x)^r}{D(x)^{r+1}} dx. \quad (7)$$

On $[0, 1]$ we have $0 \leq P(x) \leq 1/4$ and $0 < c < 1$, hence $D(x) > 0$.

We first compute two initial moments. Substituting $y = 2x - 1$ and using $\arctan(s - 1) = \pi/8$ gives

$$K_0 = \frac{4}{s} \arctan(s - 1) = s \frac{\pi}{4}. \quad (8)$$

For completeness, the arctangent identity follows from the double-angle formula: $2(s - 1)/(1 - (s - 1)^2) = 1$, and both angles lie in $(0, \pi/2)$. Next, direct differentiation gives

$$\frac{d}{dx} \frac{1 - 2x}{D(x)} = -s \frac{1}{D(x)} - 2 \frac{P(x)}{D(x)^2}.$$

The expression on the left takes the values 1 and -1 at the two endpoints. Integrating and using (8), we obtain

$$K_1 = 1 - \frac{\pi}{4}. \quad (9)$$

The same integration-by-parts mechanism gives the whole recurrence. For every $r \geq 0$, algebraic differentiation yields

$$\begin{aligned} \frac{d}{dx} \left(\frac{(1-2x)P(x)^{r+1}}{D(x)^{r+2}} \right) &= (r+1) \frac{P(x)^r}{D(x)^{r+1}} - (2r+3)s \frac{P(x)^{r+1}}{D(x)^{r+2}} \\ &\quad - 2(r+2) \frac{P(x)^{r+2}}{D(x)^{r+3}}. \end{aligned}$$

Because $P(0) = P(1) = 0$, the boundary term vanishes. Thus

$$2(r+2)K_{r+2} = (r+1)K_r - (2r+3)sK_{r+1}. \quad (10)$$

Define

$$E_r := \frac{r!s^r}{s} K_r.$$

Equations (8)–(10) become

$$E_0 = \frac{\pi}{4}, \quad E_1 = 1 - \frac{\pi}{4}, \quad E_{r+2} = (r+1)^2 E_r - (2r+3)E_{r+1}.$$

Comparing this recurrence and its two initial values with (3), induction gives the exact remainder formula

$$A_{r-1} \frac{\pi}{4} - B_{r-1} = (-1)^r E_r \quad (r \geq 0). \quad (11)$$

It remains only to show that this remainder is negligible relative to A_m . Since $P/D \leq 1/(2+s)$ on $[0, 1]$, positivity of the integrand gives

$$0 \leq K_r \leq \left(\frac{1}{2+s} \right)^r K_0.$$

Using $s/(2+s) = s-1$ and (8), this implies

$$0 \leq E_r \leq r!(s-1)^r \frac{\pi}{4}. \quad (12)$$

The positive recurrence for A_m also gives $A_{r-1} \geq r!$ by induction. Taking $r = m+1$ in (11) and applying (12), we conclude that

$$\left| \frac{\pi}{4} - \frac{B_m}{A_m} \right| \leq \frac{\pi}{4} (\sqrt{2}-1)^{m+1} \longrightarrow 0,$$

because $0 < \sqrt{2}-1 < 1$. □

Theorem 5.3 (Main). $\lim_{n \rightarrow \infty} \frac{p_n}{q_n} = \pi + e$.

Proof. Combine the exact splitting (6) with Lemma 5.1 and Theorem 5.2:

$$\frac{p_n}{q_n} = 4 \frac{B_{n+2}}{A_{n+2}} + \frac{(n+3)!}{D_{n+3}} \longrightarrow 4 \cdot \frac{\pi}{4} + e = \pi + e. \quad \square$$

Remark 5.4 (Why the problem looks hard). The four Poincaré behaviours of (1) are the products of the two $\mathcal{L}$ -behaviours (after a factorial gauge, $1 \pm \sqrt{2}$) with the two $\mathcal{D}$ -behaviours. The dominant solution is not the one carrying the answer; p_n and q_n are particular mixtures whose ratio has the two constants sitting in *different* tensor factors. Any attempt to read $\pi + e$ off the order-four recurrence directly must therefore separate a rank-two continuant factor from a rank-two factorial factor, which is what Theorem 3.1 does in closed form.

6 Formal verification

The accompanying Lean 4 development (`lean/`, toolchain `leanprover/lean4:v4.29.0`, Mathlib pinned in `lake-manifest.json`) builds with **0 errors and 0 sorry** and formalises the entire argument above, including Theorem 5.2. The principal statements are:

<code>tensor_rec</code>	Theorem 3.1, for arbitrary X, Y
<code>factorial_isDerRec</code>	Lemma 2.2
<code>challengeQ_rec, challengeP_rec</code>	the closed forms solve (1)
<code>challengeQ_zero... challengeP_three</code>	the eight initial values (2)
<code>C0_ne_zero, eq_of_satisfiesRec</code>	Lemma 4.1 and uniqueness
<code>ratio_split</code>	Corollary 4.3
<code>lambertB_div_lambertA_tendsto_pi_div_four</code>	Theorem 5.2
<code>factorial_div_derang_tendsto_exp_one</code>	Lemma 5.1
<code>problem23_pi_add_e</code>	Theorem 5.3

Every one of these depends only on the three standard axioms `propext`, `Classical.choice`, `Quot.sound`; the purely algebraic ones (`tensor_rec`, `challengeQ_rec`, `challengeP_rec`) do not even use `Classical.choice`. There is no `native_decide` anywhere. The e half is obtained from Mathlib’s `numDerangements_tendsto_inv_e` after proving that our `derang` agrees with Mathlib’s `numDerangements`.

The top-level theorem has no mathematical hypotheses:

```
theorem problem23_pi_add_e :
  Tendsto (fun m => (challengeP m : ℝ) / (challengeQ m : ℝ))
    atTop (nhds (Real.pi + Real.exp 1))
```

7 Numerical verification

The script `verify.py` performs three checks in exact arithmetic.

1. It solves (1) forward over $\mathbb{Q}$ from the initial data (2) alone and confirms that the result is integral and agrees with the closed forms (5). The first new values are $q_1 = 1836$, $q_2 = 97680$, $p_1 = 10656$, $p_2 = 573344$.
2. It verifies Theorem 3.1 symbolically, with four free initial values, so that no accidental agreement of a particular solution can be mistaken for the general identity.
3. It evaluates p_n/q_n in high precision: at $n = 50$ the agreement with $\pi + e = 5.859874482048838\dots$ is to 40 decimal places.

References

- [1] Ramanujan Machine Group, *The Ramanujan Challenge for AI*, 2026.